Enterprise Cyber Security

Joseph Maguire

Welcome Session

Overview

- Overview of Enterprise Cyber Security and what sort of topics will be explored in the course.
- Aims of the course and Intended Learning Outcomes (ILOs) of the course.
- Consider the demographic of the audience and motivation for taking the course.
- Assessment approach, the assessment design survey, and where to get help.

Course coordinator

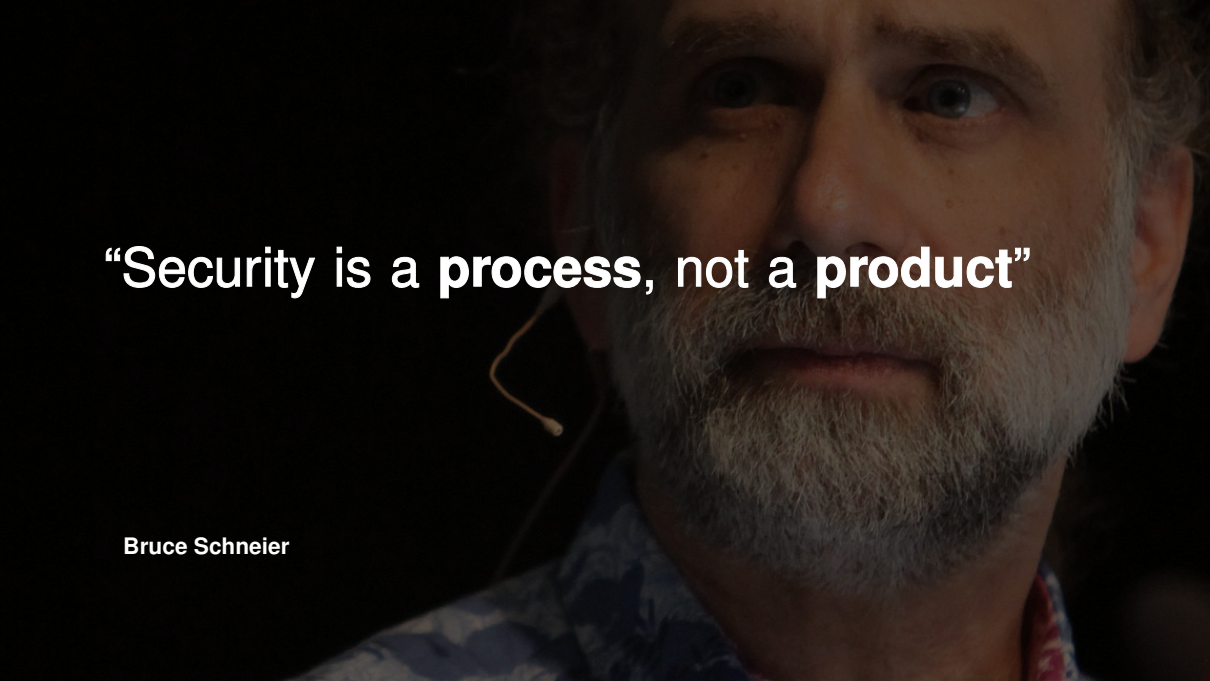
- Dr Joseph Maguire
- Email address:
  joseph.maguire@glasgow.ac.uk
- Office S112 enter via the Sir Alwyn Williams Building.
- Office hours appointments can be booked through the course Moodle.

Enterprise Cyber Security

Enterprise
Cyber Security

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Cyber Security


“Security is a **process**, not a **product**”
Bruce Schneier

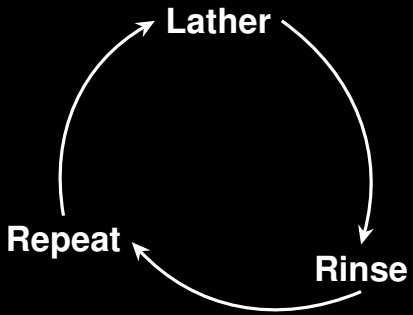
What does Schneier mean by “Security is a process, not a product”?



Answer on Mentimeter

Code

Shampoo algorithm



Security as process

- Gather data and perform an analysis to better understand the problems.
- Develop solutions to address these problems and apply them.
- Determine what success looks like and continually improve.
- Avoid continually lathering and rinsing, but never getting clean.

Tenets of cyber security

- **Prevent, detect, respond**: consider emergency services as an example.
- **People, process, technology**: security is the responsibility of everyone, not just professionals.
- **Confidentiality, integrity and availability**: information security is important to enterprises.

Course Specification

Course Specification

At a glance

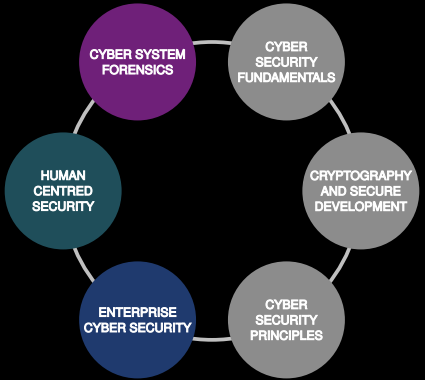
Course	Enterprise Cyber Security (M), **COMPSCI5077**
Credits	15 credits, SCQF level 11
Offered	Semester 1
Assessment	Examination **70%**, coursework **30%**
Main assessment	December

The full specification is in the University Course Catalogue.

`https://www.gla.ac.uk/coursecatalogue/course/?code=COMPSCI5077`



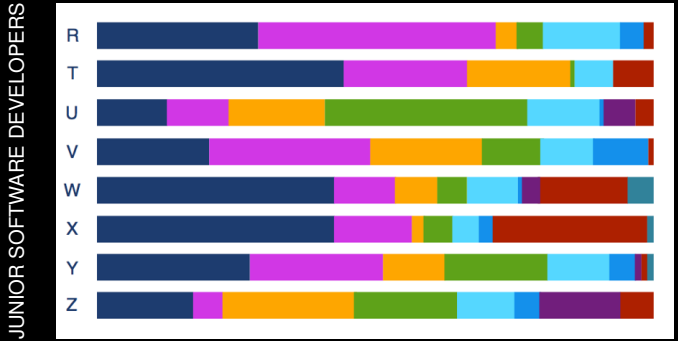
Where does the subject fit within the security options?



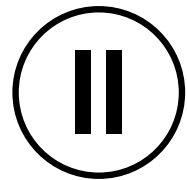
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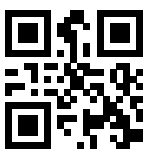
- Consider **authentication or passwords** as a cyber security process.
- **Enterprise cyber security** is primarily concerned with how the process meets the security objectives of an enterprise.
- **Cyber security fundamentals** is primarily concerned with the technical aspects of the process.
- **Human centred security** is concerned with the human challenges of using the process.

Tasks by time for graduate developers



What are the different categories?

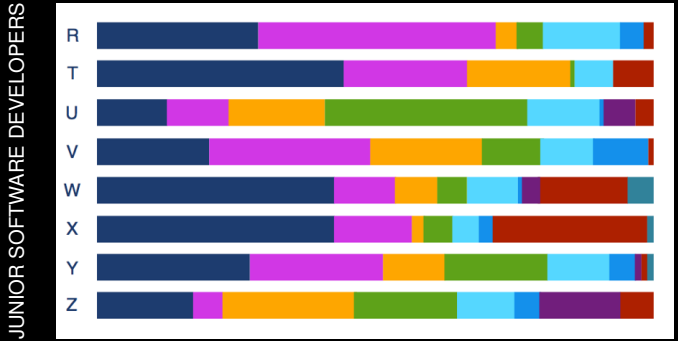




Answer on Mentimeter

Code

Tasks by time for graduate developers



Tasks by time for graduate developers



Novice Software Developers, All Over Again

Andrew Begel
Microsoft Research
1 Microsoft Way
Redmond, WA 98052
+1 (425) 705-1816
andrew.begel@microsoft.com

Beth Simon
Computer Science and Engineering Dept.
University of California, San Diego
La Jolla, CA 92093-0404
+1 (858) 534-5419
bsimon@cs.ucsd.edu

Begel, A. and Simon, B. (2008) Novice software developers, all over again. *Proceedings of ICER 2008*, ACM.

Course Aims (1)

- To develop knowledge of key concepts in **cyber security management in a business context** and an awareness of the management imperatives thereof.
- To emphasise the importance and wherewithal of information security provision to meet **legislation imperatives**.

Course Aims (2)

- To explore the advanced security management issues related to IT usage **within and outside the enterprise**, the challenges of ensuring information security and the tools available to managers to assure this.
- Encourage the acquisition of **analytical, management and communication skills** to identify enterprise assets and threats, argue mitigation actions and communicate such complex concepts to key decision makers within an enterprise.

Intended Learning Outcomes (1)

By the end of this course students will be able to:

- **Describe** different deployment concerns for a specific context.
- **Design** security policy to address perceived concerns for a specific context.
- **Argue** appropriate approaches to business continuity and resilience.
- **Predict** legal, social and ethical concerns in the secure management of information.

Intended Learning Outcomes (2)

By the end of this course students will be able to:

- **Effectively communicate** cyber security imperatives to decision makers in an organisation.
- **Identify and critically assess** threats in a specific context.
- **Critique** security policy and justification for a given context.

Which of these outcomes do you expect to find the most challenging, and why?



Answer on Mentimeter

Code

Cohort Demographic

- Individuals with knowledge and experience in **other disciplines**, but who know little of computing science.
- Individuals with **industrial insight** and experience of computing science, but little specialist knowledge.
- Individuals with **solid computing science** knowledge, but who lack specialist knowledge.
- Most will have **little to no** knowledge or experience of cyber security.

Motivation

- Gain knowledge and insight into **managing cyber security** within an enterprise context.
- Passion for computing science and the difficulties of delivering it within a challenging environment.
- Cyber security challenges touch you **every day** in terms of security and privacy.
- Generate future research and/or industrial products that have some **security thinking** to them.

Why are you taking this course, and what do you want to get out of it?



Answer on Mentimeter

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Research Reading

Linking the course to research

- Research in **enterprise and cyber security** is considered throughout the course.
- Each week you read and consider a **research paper** that extends the lecture material.
- The team exercise expects you to draw on **research and emerging thinking** when developing policy.
- The reading list is **supplementary** to lectures, not a replacement; it points to digital copies where possible.
- Reading list: `https://moodle.gla.ac.uk/mod/lti/view.php?id=6072992`



Assessment Overview

Assessment Overview

Three components

Component	Weight	In brief
Examination	70%	Main assessment in December.
Team assessed exercise	20%	Groups of 4-5; a 6-page submission to key decision makers.
Continuous assessment quizzes	10%	Take-home multiple-choice quizzes on Moodle.

Coursework (30%) is the team exercise and the quizzes together.

Team Assessed Exercise

From log evidence to security policy

- Worth **20%** of the course, groups of **4 to 5 students**.
- A fictional retailer has been attacked. Your group finds the evidence in **server logs**, assesses the risks it exposes, and designs **policy** to address the most serious.
- Communicated through a **6-page report to key decision makers**.
- **Your logs are unique to your group**: every value you report must come from your own group's logs.


LOGS
EVIDENCE
RISK ASSESSMENT
POLICY
IMPLEMENT AND EVALUATE

Team Assessed Exercise

How the report is marked

Part	Pages	Weight
1. Analysis	1.5	25%
2. Risk assessment	2	30%
3. Policy	1.5	20%
4. Implementation and evaluation	1	15%
Written quality		5%
Team performance		5%

Individual grades draw on a Workload Record and a Personal Assessment of Contribution.

Continuous Assessment

What is fixed

- Worth **10%** of the course grade.
- Take-home multiple-choice quizzes on **Moodle**.
- Run over **8 teaching weeks**; none in assessment week.
- **One attempt** per quiz.
- One mark per question, **no negative marking**.
- Only your **best 70%** of quizzes count.
- Lecture slides **and** a research paper are released every week.
- Extenuating circumstances claims apply to the **whole quiz component**.

Assessment Design Survey

Scan to respond



Assessment Design Survey

Why you have a say

- The class shapes **how the quizzes run**, through four choices.
- The aim is to give you **ownership** over the assessment, and tailor it to your motivations for the course.
- You come to appreciate the **rationale** for features of the assessment, and can explain it beyond the course itself.
- Every combination complies with the School rules for continuous assessment quizzes.


QUIZ STYLE
CADENCE
AVAILABILITY WINDOW
ATTEMPT TIME

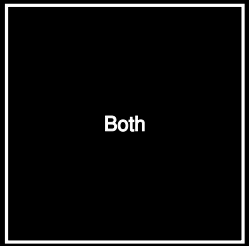
Quiz Style
Design Choices



Option A



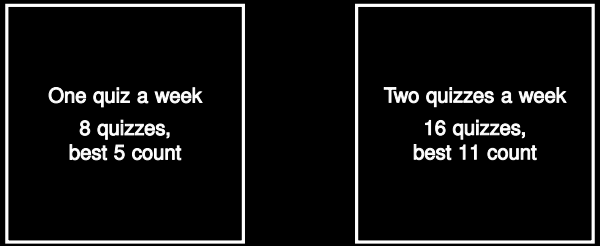
Option B



Option C


QUIZ STYLE
CADENCE
AVAILABILITY WINDOW
ATTEMPT TIME

Cadence
Design Choices



Option A

Option B

QUIZ STYLE

CADENCE

AVAILABILITY WINDOW

ATTEMPT TIME

Availability Window

Design Choices



Option A



Option B

QUIZ STYLE

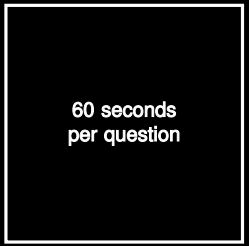
CADENCE

AVAILABILITY WINDOW

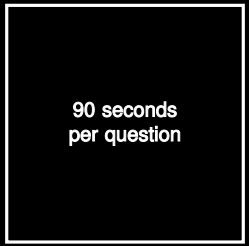
ATTEMPT TIME

Attempt Time

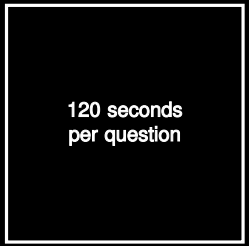
Design Choices



Option A

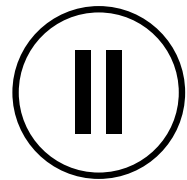


Option B



Option C

Which combination would help you learn the most, not just score the most?



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How to Vote

- Vote through the **Microsoft Form**: one question for each of the four choices.
- The form closes at **11:00 on Friday 2 October**.
- The final design is announced on Moodle **Monday 5th of October**.

Where to Get Help

Where to Get Help

On the course

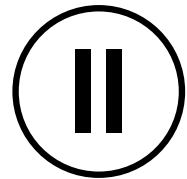
- **Course Moodle**: lecture material, the reading list, the team exercise specification, quizzes and announcements.
- **Reading list**: scan the QR code on the Research Reading slide, which links to the list.
- **Office hours**: book an appointment through the course Moodle.
- **Email**: joseph.maguire@glasgow.ac.uk
- **Your group**: raise problems within the group early, and contact me if they persist.

Where to Get Help

Across the University

- **Adviser of Studies**: questions about your programme and course choices.
- **Disability Service**: reasonable adjustments, recorded in a Study Support Plan.
- **Student Services**: wellbeing, counselling and practical support.
- **University Library**: reading list material and study skills support.

Any questions?





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